

Project Development Report

Food Survey Analysis using Tableau and Flask

The Project Development phase focused on converting the planned solution into a working application. The project integrates a Tableau dashboard with a Flask web application to analyze student food survey data. Development included dataset preparation, dashboard creation, web integration, testing, and deployment planning.

Development Methodology

The project followed an incremental approach. Each module was designed, implemented, tested, and integrated before moving to the next stage. This reduced errors and made debugging easier.

Dataset Preparation

The food_coded.csv dataset was verified for consistency and completeness. Relevant attributes such as Gender, GPA, Weight, Exercise, Healthy Feeling, Favorite Cuisine, and Indian Food Preference were organized for visualization.

Dashboard Development

Interactive worksheets were created in Tableau, including KPI cards, pie charts, histograms, scatter plots, treemaps, box plots, heat maps, and tables. These worksheets were combined into an interactive dashboard and a Tableau Story.

Flask Integration

The published Tableau dashboard was embedded into a Flask application using HTML templates. Static resources such as CSS and images were managed within the Flask project structure.

Modules Implemented

Module	Description
Dataset Module	Loads and manages the food survey CSV dataset.
Visualization Module	Generates Tableau worksheets and dashboards.
Dashboard Module	Combines visualizations into an interactive dashboard.
Story Module	Presents analytical insights in sequence.
Web Module	Displays the dashboard using Flask.
Documentation Module	Maintains reports, screenshots, and GitHub repository.

Tools and Technologies

Python, Flask, Tableau Public/Desktop, HTML, CSS, JavaScript, Git, GitHub, Visual Studio Code, and CSV datasets were used throughout development.

Challenges and Solutions

Several challenges were encountered, including dashboard layout adjustments, integrating Tableau into the Flask application, managing project files in GitHub, and organizing the repository for documentation. These were resolved by restructuring the project folders, refining the dashboard layout, testing the Flask application, and validating the published Tableau dashboard.

Development Outcomes

The final application provides an interactive dashboard that enables users to analyze food survey responses efficiently. It supports visual exploration of student eating habits, GPA distribution, healthy feeling, weight relationships, and cuisine preferences. The solution is suitable for academic demonstrations, project evaluation, and future enhancements.

Conclusion

The Project Development phase successfully transformed the project design into a functional solution. The integration of Tableau and Flask produced a user-friendly analytics platform that delivers meaningful insights from the Food Survey dataset.